

Aadiy Khan

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EDUCATION

Vellore Institute of Technology, Bhopal

2023 – 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.48/10.0

PROJECTS

FlowBERT: Customer Service Router | *Python, TensorFlow, Keras, DistilBERT*

Apr 2025 – May 2025

- Developed end-to-end AI routing system utilizing DistilBERT-based Transformer model to automate ticket categorization and priority assignment, reducing manual triage time by 75%.
- Architected Multi-Task Learning (MTL) model in TensorFlow/Keras using custom Lambda layers to simultaneously predict Department Route and Urgency Level from single text input.
- Executed data augmentation strategies scaling training dataset from 50 to 600+ records, resolving class imbalance issues and improving model robustness for edge-case customer queries.
- Deployed real-time interactive dashboard using Streamlit, enabling support teams to visualize AI-driven classifications and confidence scores for incoming service requests.

Drug Repurposing Pipeline | *RDKit, Doc2Vec, Gensim, Scikit-learn, NLTK*

Oct 2025 – Dec 2025

- Architected 5-phase drug repurposing framework integrating chemical informatics and NLP to identify novel therapeutic uses for FDA-approved compounds.
- Engineered multi-modal feature extraction system using RDKit for 2048-bit Morgan Fingerprints and Gensim (Doc2Vec) for 100-dimensional clinical embeddings, capturing molecular structure and medical intent.
- Developed high-throughput data fusion pipeline synthesizing comprehensive dataset of 2,148 features, optimizing model's ability to recognize complex drug-disease patterns.
- Implemented Direct Disease Mapping engine linking DrugBank candidates to clinical indications, utilizing Target Enrichment Analysis to rank and prioritize top-tier candidates for virtual screening.
- Automated data cleaning and preprocessing (Regex, NLTK) for thousands of drug records, ensuring high data integrity for large-scale similarity-based medical analysis.

Data Analyst Agent | *Python, Flask, Pandas, Google Gemini API, Matplotlib, Seaborn*

Jan 2026 – Feb 2026

- Engineered full-stack data analytics platform integrating Google Gemini Pro API for intelligent data insights, enabling natural language queries on uploaded datasets with automated visualization generation.
- Implemented comprehensive data preprocessing pipeline supporting CSV, Excel, and text files, featuring automated missing value handling, outlier detection using IQR method, and duplicate removal capabilities.
- Built interactive visualization engine generating custom plots (Bar, Line, Scatter, Histogram, Box) with dynamic column selection, reducing manual data exploration time by 65%.
- Deployed production-ready web application on Render with real-time AI chat interface, processing user queries and delivering statistical analysis interpretations through RESTful API architecture.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL, JavaScript, HTML, CSS

Frameworks & Libraries: TensorFlow, Keras, Transformers, Scikit-learn, Flask, Pandas, NumPy, Streamlit

Tools & Technologies: Git, Docker, RDKit, RESTful APIs, Tableau, NLTK, Gensim

Core Competencies: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Problem Solving

CERTIFICATIONS & ACHIEVEMENTS

IBM AI Engineer Professional Certificate; NPTEL Introduction to Machine Learning.

Health Hack Hackathon: Developed real-time AI routing system using DistilBERT achieving 75% reduction in manual processing time.

LeetCode: Solved 350+ algorithmic problems demonstrating strong problem-solving capabilities.

Successfully implemented the “*Attention is All You Need*” research paper from scratch.

ADDITIONAL

Languages: Fluent in Hindi, English, and Urdu

Interests: Competitive Programming, Open Source Contributions, Badminton, Football